

Quick Check

Lesson 4: Why is Lemon Juice Sour? Acids Release H^+ Ions in Water (Arrhenius Definition)

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.1: Acids and Bases | Lesson 4

Pure water has pH 7.0 and H^+ concentration of $1 \times 10^{-7} \text{ mol/L}$. Which statement about the H^+ ions in pure water is correct?

A It has a tiny amount of H^+ , far less than lemon juice

B It has more H^+ than vinegar

C It has the same amount of H^+ as lemon juice

D It has no H^+ ions at all

Lemon juice has pH 2.0 and $H^+ = 1 \times 10^{-2}$ mol/L. Vinegar has pH 2.8 and H^+ about 1.6×10^{-3} mol/L. About how many times more H^+ does lemon juice have than vinegar?

A About 0.8 times

B About 10 times

C About 6 times

D About 60 times

In a particle simulation, acid molecules from lemon juice dissolve in water. What do learners see happen?

- A** The acid molecules stay whole and release nothing
- B** The acid molecules turn into sugar particles
- C** The acid molecules remove all the water particles
- D** The acid molecules release H^+ (aq) ions into the water

Which statement is the Arrhenius definition of an acid?

- A** A substance that is harmful if swallowed
- B** A substance that contains no water
- C** A substance that produces H^+ (aq) ions when dissolved in water
- D** A substance that tastes sour in the laboratory

Lemon juice (pH 2.0) and vinegar (pH 2.8) are both acids. Why does lemon juice have the lower pH?

- A** It contains no water, but vinegar does
- B** It has fewer H^+ ions than vinegar
- C** It has a higher concentration of H^+ ions than vinegar
- D** It contains more sugar than vinegar

Stomach acid is hydrochloric acid. Which particle does it release when it dissolves in water, making it an acid?

A No particles, because it is a liquid

B Only water particles

C H^+ (aq) ions

D Sugar particles

In a Model drawing, lemon juice ($\text{H}^+ = 1 \times 10^{-2} \text{ mol/L}$) is shown with 6 H^+ dots. Which drawing for vinegar (about $1.6 \times 10^{-3} \text{ mol/L}$) and water ($1 \times 10^{-7} \text{ mol/L}$) is best?

- A Vinegar 6 dots, water 0 dots
- B Vinegar 1 dot, water almost no dots
- C Vinegar 1 dot, water exactly 0 dots
- D Vinegar 12 dots, water 6 dots

Lemon juice turns blue litmus paper red. Which explanation uses the idea from this lesson?

A Lemon juice contains H^+ (aq) ions released by its acid in water

B Lemon juice is red in colour

C Lemon juice has no water particles

D Lemon juice contains sugar particles

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).